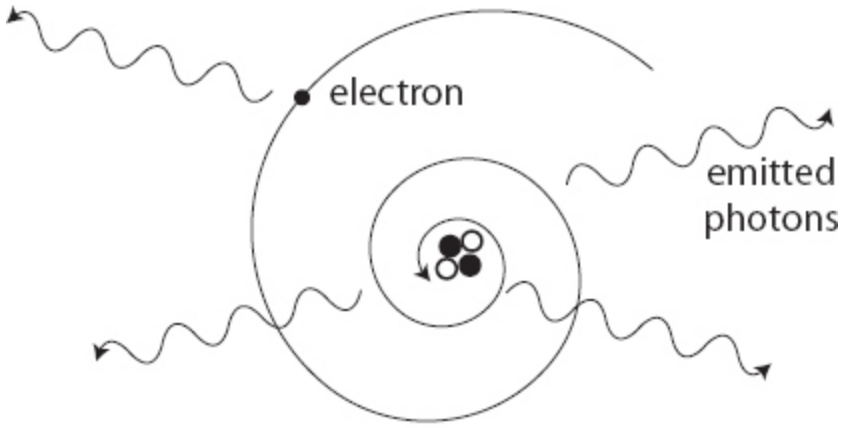


Instability of the classical Rutherford atom

Perhaps all of your atoms are giving off light, but it's just too faint to see. After all, identical logic applies to the planets in the solar system. They should be giving off gravitational waves—an accelerating mass should cause ripples in the gravitational field, just like an accelerating charge causes ripples in the electromagnetic field. And indeed they are. If there was any doubt that this happens, it was swept away in 2016, when researchers at the LIGO and Virgo gravitational-wave observatories announced the first direct detection of gravitational waves, created when black holes over a billion light years away spiraled into each other.

But the planets in the solar system are much smaller, and move more slowly, than those black holes, which were each over thirty times the mass of the sun. As a result, the emitted gravitational waves from our planetary neighbors are very weak indeed. The power emitted in gravitational waves by the orbiting Earth amounts to about 200 watts—equivalent to the output of a few lightbulbs, and completely insignificant compared to other influences such as solar radiation and tidal forces. If we pretend that the emission of gravitational waves were the only thing affecting the Earth's orbit, it would take over 10^{23} years for it to crash into the sun. So perhaps the same thing is true for atoms: maybe electron orbits aren't really stable, but they're stable enough.

This is a quantitative question, and it's not hard to plug in the numbers and see what falls out. The answer is catastrophic, because electrons should move much faster than planets and electromagnetism is a much stronger force than gravity. The amount of time it would take an electron to crash into the nucleus of its atom works out to about ten picoseconds. That's one-